

Name: _____





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There Was an Old Lady Who Swallowed a Fly

Corpus

There was an old lady who swallowed a fly,
I don't know why she swallowed a fly – perhaps she'll die!

There was an old lady who swallowed a spider
That wriggled and jiggled and tickled inside her;
She swallowed the spider to catch the fly;
I don't know why she swallowed a fly – perhaps she'll die!

There was an old lady who swallowed a bird;
How absurd to swallow a bird!
She swallowed the bird to catch the spider
That wriggled and jiggled and tickled inside her,
She swallowed the spider to catch the fly;
I don't know why she swallowed a fly – perhaps she'll die!

There was an old lady who swallowed a cat;
Well, fancy that, she swallowed a cat!
She swallowed the cat to catch the bird,
She swallowed the bird to catch the spider
That wriggled and jiggled and tickled inside her,
She swallowed the spider to catch the fly;
I don't know why she swallowed a fly – perhaps she'll die!

Chunking the Title

1) How many 1-, 2-, and 3-word chunks can you find in the first line of the song? Finish the list of chunks in each row.

Words	Chunks	Count
1 word	"there", "was", "an", "old", "lady", "who", "swallowed", "a", "fly" ...	
2 words	"there was", "was an", "an old", "old lady" ...	
3 words	"there was an", "was an old", "an old lady" ...	

2) In this corpus (**the lyrics and the title!**) How many times does the word "there" appear? _____

3) In this corpus, how many times was the word "there" followed by the word "was"? _____

4) If you picked any of the "there"s in the corpus, what is the probability that the next word would be "was"? _____

Constructing a Language Model

Use [the corpus](#) to answer the questions below. Remember: the title of the song is a part of the corpus!

The example below shows the probability that "was" will come after "there".

$$p(\text{was} | \text{there}) = \frac{\text{count}(\text{there was})}{\text{count}(\text{there...})} = \frac{5}{5}$$

Did she swallow a fly?

Let's say we want to know the likelihood that the old lady "swallowed a fly". We need a ratio!

1) $p(\text{fly} | \text{swallowed a}) = \frac{\text{count}(\text{swallowed a fly})}{\text{count}(\text{swallowed a...})} = \frac{\quad}{10}$

2) The likelihood that the old lady "swallowed a fly" in the excerpt is _____.

What about the other animals?

Find the ratio representing the likelihood that she swallowed each of the other animals.

3) $p(\text{spider} | \text{swallowed a}) = \frac{\text{count}(\text{swallowed a spider})}{\text{count}(\text{swallowed a...})} = \frac{\quad}{10}$

4) $p(\text{bird} | \text{swallowed a}) = \frac{\text{count}(\text{swallowed a bird})}{\text{count}(\text{swallowed a...})} = \frac{\quad}{10}$

5) $p(\text{cat} | \text{swallowed a}) = \frac{\text{count}(\text{swallowed a cat})}{\text{count}(\text{swallowed a...})} = \frac{\quad}{10}$

6) Across the entire poem, the lady *most likely* swallowed a _____.

7) The second most likely option is that she swallowed a _____.

8) There is an equal probability that she swallowed a _____ or a _____.

To catch the...?

Find the ratio representing the likelihood that she caught each of the animals.

9) $p(\text{fly} | \text{to catch the}) = \frac{\text{count}(\text{to catch the fly})}{\text{count}(\text{to catch the...})} = \frac{\quad}{6}$

10) $p(\text{spider} | \text{to catch the}) = \frac{\text{count}(\text{to catch the spider})}{\text{count}(\text{to catch the...})} = \frac{\quad}{6}$

11) $p(\text{bird} | \text{to catch the}) = \frac{\text{count}(\text{to catch the bird})}{\text{count}(\text{to catch the...})} = \frac{\quad}{6}$

12) The lady was **most** likely trying to catch the _____. It is the **least** likely that she was trying to catch the _____.

The...?

13) $p(\text{fly} | \text{the}) = \frac{\quad}{\quad} = \frac{\quad}{\text{decimal}} = \frac{\quad}{\text{percent}}$

14) $p(\text{spider} | \text{the}) = \frac{\quad}{\quad} = \frac{\quad}{\text{decimal}} = \frac{\quad}{\text{percent}}$

15) $p(\text{bird} | \text{the}) = \frac{\quad}{\quad} = \frac{\quad}{\text{decimal}} = \frac{\quad}{\text{percent}}$

16) $p(\text{cat} | \text{the}) = \frac{\quad}{\quad} = \frac{\quad}{\text{decimal}} = \frac{\quad}{\text{percent}}$

Sampling from the Language Model

Use Pyret to answer the questions below.

```
# build-lang-model :: ( String ) -> Table
                        corpus

# generate-ngrams :: ( Table , Number ) -> Table
                        model    size

# completions :: ( Table , String ) -> Table
                  model    start
```

Text Generation Exercise #1

1) Let's start our text generation with the phrase "She ".

Text Generation 1: She
 first word second word third word fourth word

2) Determine which word is most likely to follow "She " and record it on the line for second word above.

3) Determine which word is most likely to follow the word you just wrote down and record it on the line for third word above.

4) Use statistical language modeling to determine the fourth word.

5) Everyone in your class should have generated the **same** text. Why do you think that was the outcome?

Text Generation Exercise #2

Here is a list of the most common unigrams in the corpus: "the": 12 times "she": 11 times "a": 11 times

6) Let's start by choosing the **most** common word:

7) Determine which word is most likely to follow that word:

8) There are two words that have an equal probability of appearing in the third spot! What are they? and

9) Use statistical language modeling to determine the fourth word for each of the sentences below, using each of the two words you computed for the third spot.

Text Generation 2a: the
 first word second word third word fourth word

Text Generation 2b: the
 first word second word third word fourth word

10) Why was there only one result for **Text Generation Exercise 1**, while **Text Generation Exercise 2** had two possible results?

★ What is another starting word that would result in only one possible four-word phrase?

★★ What is another starting word that would result in at least two possible outcomes?

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